

BUILDTRACK SYSTEM SPECIFICATION

Comprehensive Backend Architecture, End-to-End Workflows, and Complete REST API Reference

Node.js Express 5.2 Google Firebase Firestore Flutter Mobile Client React 19 Admin Web 66 Active Endpoints

1. BACKEND ARCHITECTURE & TECH STACK

Component	Technology	Role & Implementation Details
Server Framework	Express.js v5.2.1 (Node.js)	Monolithic REST API server handling request validation, routing, JSON payload parsing, and CORS policies.
Primary Database	Google Firebase Firestore	NoSQL Cloud Document Database managed through <code>firebase-admin v14.5.0</code> service account credentials.
Security & Encryption	bcrypt v6.0.0	One-way salted password hashing for both Admin web accounts and Worker mobile profiles.
Session Tokens	jsonwebtoken (JWT) v9.0.3	Stateless token authentication for secure admin sessions and password reset authorization.
Notification Engine	Nodemailer v10.0.13	SMTP email transport service used for automated admin password recovery verification codes.
Large Payload Support	express.json (Limit: 50MB)	Supports high-resolution receipt image uploads, base64 avatars, and OCR transaction attachments.

2. DATABASE SCHEMA (11 FIRESTORE COLLECTIONS)

projects Project name, client, location, original budget, timeline, status, and milestone metadata.	budget_additions History log of project budget top-ups, approved amounts, reasons, and timestamps.
expenses Financial transaction logs, receipt image attachments, categories, amounts, and approvals.	workers Worker profiles, QR IDs, position keywords (Worker, Attendance Monitoring, Tools Monitoring), approval status.
attendance Daily field time-in/out records, worker IDs, geo/site details, overtime, and attendance status.	assets Tool & equipment registry, current status (Available, In Use, Maintenance), and condition grades.
borrow_history Complete immutable audit trail of all check-outs and check-ins, custodians, and return conditions.	materials Raw materials inventory per project, current stock levels, units, and minimum re-order thresholds.
cash_advances Worker cash advance (bale) requests, requested amount, repayment balance, and admin approval status.	schedules Project timeline tasks, equipment maintenance dates, and calendar events linked to projects.
admins Web portal administrator profiles, encrypted credentials, system privileges, and preferences.	notifications System alerts triggered by low material stock, unreturned tools, or pending worker sign-ups.

3. END-TO-END WORKFLOW ARCHITECTURE

A. User Onboarding & Access Control

- **Admin Web:** Direct secure login via `/api/admin/login` with session tokens. Includes email-based password recovery via Nodemailer.
- **Mobile Workers:** Self-registration on Flutter app enters a *Pending* approval state. Admin approves worker profiles on Web Dashboard before login access is granted.

B. Project Creation & Dynamic Budget Allocation

- Admin registers projects on Web Portal with base capital allocation.
- Live project expense tracking deducts from budget in real time.
- Additional project capital injections are recorded via `/api/projects/:id/budget-additions` for transparent financial auditing.

C. Expense Tracking & OCR Receipt Integration

- Site engineers and staff log material and logistical costs.
- Receipts can be captured directly or scanned via client-side OCR for automated field filling.
- Aggregated financials are served via monthly and category summary endpoints for analytics charts.

D. Attendance Monitoring & On-Site QR Verification

- Personnel designated with *Attendance Monitoring* position use the Mobile App to scan worker QR identification.
- Time-in, Time-out, and late calculations sync instantly to Firestore.
- Admin monitors real-time headcounts, present lists, and manpower costs on the Web Portal.

E. Equipment Custody & Tool Checkout Pipeline

- **Web Portal (Admin):** Strictly handles tool inventory creation, editing, maintenance logs, and monitoring unreturned tools.
- **Mobile App (Field Custodian):** Exclusively manages tool Check-Out on-site.
- Returns update tool condition (Good / Needs Repair / Broken) and automatically archive to `borrow_history`.

F. Worker Cash Advance (Bale) Lifecycle

- Workers submit cash advance requests via Flutter Mobile App with specified amounts and reasons.
- Requests surface on Admin Web Portal under *Pending Advances*.
- Admin approval/rejection updates worker balance ledger and logs the transaction.

4. COMPLETE REST API ENDPOINTS SPECIFICATION (66 ENDPOINTS)

A. Authentication & Admin Management (10 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
POST	/api/admin/login	Authenticates administrator credentials and returns session data.
POST	/api/admin/register	Registers a new administrator profile in Firestore.
POST	/api/admin/forgot-password	Generates password reset token and emails it via Nodemailer.
POST	/api/admin/reset-password	Validates reset token and sets new encrypted password.
GET	/api/admins	Retrieves list of all registered administrators.
GET	/api/admin/profile/:id	Fetches profile details, contact, and preferences for an admin.
POST	/api/admin/profile/save	Updates admin personal details and avatar image.
PUT	/api/admin/:id/password	Updates admin password from the settings dashboard.
PUT	/api/admin/:id/settings	Saves system preferences and notification configurations.
GET	/api/pending-registrations	Fetches worker registrations awaiting admin approval.

B. Worker Authentication & Management (8 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
POST	/api/register	Worker sign-up from mobile app (status default: Pending).
POST	/api/login	Worker mobile login (authenticates username or phone number).
GET	/api/workers	Lists all registered workers with approval status.
GET	/api/workers/summary	Aggregates active, pending, and total manpower counts.
POST	/api/workers	Admin directly creates a worker profile with assigned positions.
PUT	/api/workers/:id	Updates worker position (supports multi-role comma separated values).
DELETE	/api/workers/:id	Deletes worker profile record.
PUT	/api/workers/:id/approve	Admin approves pending mobile registration to grant login access.

C. Project & Budget Management (6 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/projects	Fetches all active and completed construction projects.
POST	/api/projects	Creates a new project record with initial budget.
PUT	/api/projects/:id	Updates project timelines, location, or status.
DELETE	/api/projects/:id	Removes project and associated data references.
GET	/api/projects/:id/budget-additions	Retrieves audit log of additional funds injected into project.
POST	/api/projects/:id/budget-additions	Adds additional budget top-up to active project.

D. Expenses & Financial Accounting (8 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/expenses	Retrieves all global financial transactions.
GET	/api/expenses/:project_id	Retrieves filtered expenses for a specific project.
GET	/api/expenses/monthly	Aggregates expenses by month for analytical trend charts.
GET	/api/expenses/budget-summary	Calculates total budget, total spent, and remaining variance.
GET	/api/expenses/summary	Returns expense totals grouped by operational category.
POST	/api/expenses	Logs new expense receipt with amount, category, and date.
PUT	/api/expenses/:id	Edits existing expense transaction record.
DELETE	/api/expenses/:id	Deletes expense record and updates budget tally.

E. Attendance & Timekeeping (4 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/attendance/:project_id	Fetches attendance logs for a project (resolves project aliases).
POST	/api/attendance	Logs attendance entry (time-in/out) via Mobile App QR scan.
PUT	/api/attendance/:id	Updates recorded attendance record details.
PUT	/api/attendance/:id/status	Updates status flag (Present, Late, Half-Day, Overtime).

F. Equipment, Tools (Assets) & Borrow History (9 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/assets	Retrieves full equipment inventory across all projects.
GET	/api/assets/:project_id	Retrieves equipment assigned to a specific project.
GET	/api/assets/summary	Summary counts of Available, In-Use, and Maintenance tools.
POST	/api/assets	Registers new tools into inventory (supports batch insertion).
POST	/api/assets/:project_id	Registers new tool directly bound to a project.
PUT	/api/assets/:id	Updates tool status or condition (Good, Needs Repair, Broken).
DELETE	/api/assets/:id	Removes tool asset from inventory.
GET	/api/borrow-history/:project_id	Fetches historical checkout and check-in logs for a project.
POST	/api/borrow-history	Records a checkout or check-in transaction into audit log.

G. Materials & Inventory Control (5 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/materials/:project_id	Lists all material stocks recorded for a project.
POST	/api/materials	Logs new material shipment with quantity and unit.
POST	/api/materials/:project_id	Adds material tied directly to project scope.
PUT	/api/materials/:id	Updates stock level, status, or unit quantity.
DELETE	/api/materials/:id	Removes material entry from database.

H. Cash Advance / Bale System (5 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/cash-advances/pending	Retrieves all outstanding cash advance requests awaiting review.
GET	/api/cash-advances/:project_id	Retrieves cash advances filed under a project.
GET	/api/workers/:id/cash-advances	Retrieves personal cash advance history for a worker.
POST	/api/cash-advances	Worker files a new cash advance request from mobile app.
PUT	/api/cash-advances/:id/approve	Admin approves or rejects cash advance request.

I. Schedules & Project Milestones (4 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/schedules	Fetches calendar milestones with project/expense links.
POST	/api/schedules	Creates a schedule item linked to a project or expense.
PUT	/api/schedules/:id	Updates scheduled date, assignees, or completion state.
DELETE	/api/schedules/:id	Deletes milestone from project calendar.

J. System Reports & Business Intelligence (4 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/reports/expenses	Consolidated financial audit report with category breakdown.
GET	/api/reports/manpower	Workforce metrics, attendance rates, and labor cost summaries.
GET	/api/reports/materials	Material utilization and remaining inventory audit report.
GET	/api/reports/assets	Equipment usage, maintenance costs, and active inventory report.

K. Notifications Engine (2 Endpoints)

METHOD	ENDPOINT	DESCRIPTION / PURPOSE
GET	/api/notifications	Retrieves active system alerts and operational notifications.
PUT	/api/notifications/:id/read	Marks individual notification alert as read.